ERDOS RESEARCH

POSITION PAPER

Learning Machine Learning and Deep Learning

through the Lens of Measure Theory

A Pedagogical and Research Position
for the Foundations of AI Curriculum

Ajit Jaokar
Erdos Research
April 2026

Abstract

This position paper argues that machine learning and deep learning are most fruitfully understood not as a collection of architectures and optimisation tricks, but as a unified theory of operations on probability measures. The dominant pedagogy — function approximation by gradient descent on labelled data — succeeds at producing engineers who can train models, but produces practitioners who cannot reason about why algorithms fail, why divergences are not interchangeable, or why the next generation of generative models looks the way it does. We propose a curriculum and research stance built on five foundational measure-theoretic objects (the data measure, the model measure, the pushforward, divergences, and the Radon-Nikodym derivative) and four operational modes (density estimation, conditional estimation, distribution matching, and sampling). We argue that this framework unifies VAEs, GANs, normalising flows, diffusion models, and energy-based models within a single conceptual structure; predicts failure modes from preconditions; and connects machine learning to the mature mathematical disciplines (optimal transport, stochastic analysis, information geometry, ergodic theory) that increasingly drive frontier research. We close with concrete recommendations for course design, prerequisite ordering, and assessment.

Machine learning is the problem of approximating, transforming, and sampling from probability measures using finite samples drawn from those measures.

1. The Problem with How We Teach Machine Learning

The standard introductory treatment of machine learning begins with linear regression as a curve-fitting exercise, progresses through logistic regression as a classification boundary, introduces neural networks as universal function approximators, and arrives at deep learning as a sequence of architectural innovations: convolutions, attention, transformers. Probability enters through the back door — usually as a justification for cross-entropy loss or as a story about Bayesian inference appended to a chapter on regularisation. Measure theory, when it appears at all, appears in graduate-level appendices for students who have already absorbed the engineering vocabulary.

This ordering produces a particular kind of practitioner. They can configure a model, tune hyperparameters, and read a research paper. What they cannot reliably do is the following:

- Explain why maximum likelihood training fails when the data lives on a low-dimensional manifold inside a high-dimensional ambient space.
- Predict, before running the experiment, that a vanilla GAN will exhibit vanishing gradients when the generator and data distributions have disjoint support.
- Articulate the precise reason that importance-weighted covariate shift correction breaks down when the source and target distributions are nearly singular with respect to one another.
- Connect the score function in diffusion models to the Radon-Nikodym derivative of one path-space measure with respect to another.
- Choose between Wasserstein, KL, Jensen-Shannon, and Maximum Mean Discrepancy with reasoning grounded in the structure of the problem rather than empirical folklore.

These are not edge cases. They are the substance of contemporary research in generative modelling, distribution shift, and probabilistic inference. The gap between what students learn and what the frontier requires is not a gap in architectural knowledge or compute — it is a gap in the language of measures.

1.1 The Engineering Dialect and Its Limits

There is nothing wrong with engineering pedagogy as such. Practitioners need to ship working systems, and abstraction without grounding produces a different and equally limited kind of student. The problem is that the engineering dialect of machine learning has reached the limits of what it can express. When researchers describe a diffusion model as the time-reversal of a stochastic differential equation, when flow matching is presented as an optimal transport interpolation, when score-based generative modelling is formulated through the Fokker-Planck equation — these are not optional decorations on top of an engineering core. They are the core, and the engineering description is now the decoration.

A curriculum that delays measure theory until a graduate appendix forces students to relearn the field once they encounter modern generative modelling. A curriculum that introduces

measure theory at the foundation gives them the vocabulary to absorb new architectures as instances of operations they already understand.

1.2 What This Paper Argues

This paper argues three things. First, that the central objects of machine learning are probability measures, and the central operations are estimation, transformation, divergence minimisation, and sampling. Second, that this reframing is not a notational re-dressing but a substantive change in what students learn to reason about. Third, that the reframing yields a curriculum that is more unified, predicts failure modes more reliably, and connects more cleanly to the mathematical disciplines that drive contemporary research.

The remainder of the paper develops the framework in detail (Section 2), demonstrates its explanatory power by mapping major algorithm families onto it (Section 3), examines the failure-mode predictions it makes (Section 4), addresses anticipated objections (Section 5), and proposes a concrete curriculum structure (Section 6).

2. The Framework

We organise the framework around five foundational objects and four operational modes. The five objects are the vocabulary; the four modes are the grammar. Every algorithm in machine learning is a sentence built from these elements.

2.1 The Five Foundational Objects

The Data Measure

There is an unknown probability measure P_data defined on a measurable space (X, Σ_X). This measure represents the true distribution from which observations are drawn. It is unknown not in the sense of being slightly imprecise, but in the sense of being epistemically inaccessible: we never observe P_data directly. We observe finite samples.

The empirical measure P_n is the discrete probability measure that places mass 1/n on each of n observed points. It is the object that actually enters our computations. The fundamental epistemic problem of machine learning is that P_n is discrete while P_data is typically continuous, or supported on a low-dimensional submanifold of high-dimensional ambient space — the manifold hypothesis. Every learning algorithm is, in some way, a strategy for inferring properties of P_data from properties of P_n.

The Model Measure

The model measure P_θ is a parametric family of probability measures, indexed by parameters θ in some parameter space Θ. Training is search over Θ. The phrase "the model learns the distribution" is shorthand for "the optimisation procedure selects a θ such that P_θ is a good approximation, in some specified sense, of P_data." The phrase "in some specified sense" is doing enormous work and is the subject of Mode 3 below.

The Pushforward Measure

Given a measurable map $f: X \to Y$ and a measure μ on X, the pushforward measure $f^*\mu$ on Y is defined by $(f^*\mu)(A) = \mu(f^{-1}(A))$ for measurable $A \subset Y$. The pushforward is what a neural network does to a distribution. A network is a measurable map; passing samples through it transports the input distribution to an output distribution. Forward passes on minibatches are Monte Carlo estimates of pushforwards. Generative models, viewed measure-theoretically, are pushforwards of a simple base measure (typically a Gaussian) through a learnable map.

The Divergence

A divergence D(P || Q) is a functional on pairs of probability measures that quantifies dissimilarity. Divergences are not interchangeable. The Kullback-Leibler divergence requires P to be absolutely continuous with respect to Q; when this condition fails, KL is infinite. The Jensen-Shannon divergence is bounded but has zero gradient where the supports of P and Q are disjoint. The Wasserstein-1 distance is well-defined for measures with disjoint support and metrises weak convergence, but requires solving an optimal transport problem. Maximum Mean Discrepancy admits closed-form unbiased estimators from samples, but depends on a kernel choice. The total variation distance bounds error probability uniformly

over events but is hard to estimate from samples. Each of these properties has direct algorithmic consequences. The choice of divergence is the choice of what kind of approximation one obtains.

The Radon-Nikodym Derivative

When P is absolutely continuous with respect to Q, written $P \ll Q$, the Radon-Nikodym derivative dP/dQ exists and is the rigorous object behind likelihood ratios, importance weights, and density ratio estimation. When P is not absolutely continuous with respect to Q — for example, when one is supported on a manifold and the other on the ambient space — the derivative does not exist, and any algorithm that implicitly assumes its existence will fail. This single fact explains a significant fraction of the failure modes catalogued in Section 4.

2.2 The Four Operational Modes

Every machine learning algorithm performs one or more of the following operations on measures.

Mode 1 — Density Estimation

The goal is to find P_θ that approximates P_{data} directly. Maximum likelihood training is exactly minimisation of the Kullback-Leibler divergence $KL(P_{data} \,||\, P_\theta)$, which requires $P_{data} \ll P_\theta$. This precondition is non-trivial; it fails when P_{data} is supported on a manifold and P_θ has full-dimensional support, and it fails in the opposite direction as well. Algorithms in this mode include autoregressive models, normalising flows, variational autoencoders (which optimise a tractable lower bound because the marginal is intractable), energy-based models, and diffusion models (which factorise density estimation through a sequence of progressively easier denoising problems).

Mode 2 — Conditional Estimation

Supervised learning estimates the conditional measure $P_{data}(Y \mid X)$, itself a measure-theoretic object whose existence as a regular conditional probability requires conditions that are usually satisfied in practice (Y in a Polish space) but worth stating. Cross-entropy loss is the conditional KL divergence. Mean squared error is the squared $L^2(P_X)$ norm, in which the conditional expectation $E[Y \mid X]$ is the orthogonal projection onto the closed subspace of $\sigma(X)$-measurable functions. Class imbalance, importance weighting, and covariate shift correction are all reweightings of underlying measures, with the Radon-Nikodym derivative dP_{target}/dP_{source} playing the role of the importance weight.

Mode 3 — Distribution Matching

Here the goal is to minimise a divergence between P_θ and P_{data} without necessarily representing P_θ as a tractable density. Generative adversarial networks minimise Jensen-Shannon (in the original formulation) or Wasserstein-1 (in WGAN). Score matching minimises the Fisher divergence. MMD-based generative models minimise Maximum Mean Discrepancy via kernel embeddings into a reproducing kernel Hilbert space. The structural

insight here is that the choice of divergence determines what kind of approximation one is computing, and the trade-offs are not aesthetic preferences but mathematical consequences.

Mode 4 — Sampling

Some algorithms do not represent their target measure tractably; they only sample from it. Markov chain Monte Carlo samples from a Boltzmann measure proportional to exp(-E(x)). Langevin dynamics samples from any measure with a known score $\nabla \log p$. Diffusion sampling integrates a reverse-time stochastic differential equation whose stationary measure is P_data. Bayesian neural networks sample from posteriors via Hamiltonian Monte Carlo or variational approximations. This mode is where machine learning meets stochastic processes on measure spaces, and where the path-space view of generative modelling becomes essential.

3. Unification: Major Algorithm Families on a Single Map

The framework's first claim — that disparate algorithms are instances of common operations — is best demonstrated by mapping. The table below places twelve algorithm families on a single grid of (mode, divergence, regularity required, failure mode). This table is the central pedagogical artefact of the framework: students who can complete the rows for new algorithms have internalised the conceptual structure.

Algorithm	Mode	Divergence / Loss	Regularity Required	Failure Mode
Linear / Logistic Regression	Conditional	L^2 / Cond. KL	$E[\|Y\|^2] < \infty$ for L^2; finite KL for classification	Misspecification when $P(Y\|X)$ is far from the parametric family
Autoregressive Models	Density	KL (MLE)	$P_{data} \ll P_\theta$; chain-rule factorisation valid	Order sensitivity; exposure bias under teacher forcing
Normalising Flows	Density	KL (MLE)	$P_\theta \ll$ Lebesgue; tractable Jacobian; bijective f	Cannot model manifold-supported data; topology constraints on f
Variational Autoencoders	Density (bound)	ELBO = -KL + recon	Tractable variational family $q(z\|x)$	Posterior collapse; blurry reconstructions from amortisation gap
Vanilla GAN	Distribution Matching	Jensen-Shannon	Overlapping supports of P_g and P_{data}	Vanishing gradients on disjoint supports; mode collapse
WGAN	Distribution Matching	Wasserstein-1	1-Lipschitz critic (Kantorovich-Rubinstein duality)	Lipschitz enforcement (clipping vs gradient penalty) is delicate
Score Matching	Density (implicit)	Fisher divergence	$\nabla \log p_{data}$ exists in $L^2(P_{data})$; smoothness of density	Score is undefined on manifolds without noise perturbation
Diffusion Models	Sampling + Density	Denoising score matching	Forward SDE has tractable transition kernel	Discretisation error in reverse SDE; conditioning fragility
Energy-Based Models	Density (unnormalised)	Contrastive / score	Partition function finite; MCMC mixing	Intractable normalisation; slow MCMC convergence
MMD-GAN / Kernel Methods	Distribution Matching	MMD (RKHS)	Characteristic kernel; bounded second moment	Kernel choice and bandwidth determine what is matched
Bayesian Neural Networks	Sampling	Posterior KL (VI) or MCMC	Prior $\ll$ posterior; ergodicity for MCMC	Variational gap; HMC tuning; cold posterior effect

Algorithm	Mode	Divergence / Loss	Regularity Required	Failure Mode
Flow Matching	Sampling	Vector field regression	Continuous-time interpolant between base and target measure	Choice of interpolant affects sampling efficiency

Several observations follow from the table. First, the regularity column captures preconditions that determine where an algorithm can be applied. Normalising flows require absolute continuity with respect to Lebesgue measure, which is why they cannot model the manifold-supported distributions that GANs and diffusion models handle naturally. Second, the failure-mode column is not an empirical catalogue of difficulties encountered in practice; each entry is a direct consequence of the regularity requirement. Third, algorithms in the same mode but with different divergences (vanilla GAN vs WGAN, KL-based MLE vs MMD) are doing fundamentally different things, even when they produce superficially similar samples. The framework makes this difference explicit.

4. Prediction: What the Framework Tells You Before You Train

A pedagogical framework is valuable in proportion to the predictions it licenses. The measure-theoretic framework yields a small number of predictions that are non-obvious to students trained in the engineering dialect, and that are vindicated repeatedly across the contemporary literature. We give five.

4.1 KL Divergence Fails on Manifolds

Real data — natural images, audio, text embeddings — is widely believed to lie on or near low-dimensional submanifolds of high-dimensional ambient space. The Lebesgue measure of any such submanifold is zero. Consequently, a model P_θ with full-dimensional support is mutually singular with respect to a manifold-supported P_{data}, and $KL(P_{data} \| P_\theta)$ is infinite. Maximum likelihood is in this sense ill-posed for natural data, and the apparent success of MLE-based models depends on smoothing assumptions (added noise, density on the ambient space) that the framework makes visible. Students who understand this can predict, before encountering them, that score-based and diffusion models — which add noise precisely to restore absolute continuity — would emerge.

4.2 Vanishing Gradients in GANs Are Not a Tuning Problem

When generator and data distributions have disjoint or nearly disjoint supports, the Jensen-Shannon divergence is constant (equal to $\log 2$), and its gradient with respect to generator parameters is zero. This is not a bug in optimisation; it is a property of the divergence. The Wasserstein distance, by contrast, varies smoothly with the distance between supports. The transition from JS-GANs to WGANs in the research literature is a direct consequence of this structural fact, and the framework allows students to anticipate it rather than absorb it as historical accident.

4.3 Importance Weighting Breaks When Densities Are Nearly Singular

Covariate shift correction multiplies losses by the ratio dP_{target}/dP_{source}. When this ratio is unbounded — which occurs whenever the target places mass where the source places little — the importance-weighted estimator has unbounded variance and fails to converge at standard rates. The framework predicts this directly from the integrability conditions on the Radon-Nikodym derivative. The widely observed failure of naive importance weighting in deep learning settings is a precise consequence.

4.4 Score Functions Are Undefined Without Smoothing

The score $\nabla \log p(x)$ is undefined where the density is zero, and undefined as a tempered distribution where the density is supported on a measure-zero set. Adding noise (as in denoising score matching, diffusion models, and noise-conditional score networks) is not a regularisation trick — it is the minimal modification that restores the existence of the object the algorithm needs to estimate. Students who internalise this can read a new score-based paper and immediately identify what role noise is playing.

4.5 Posteriors in High Dimensions Concentrate on Unexpected Sets

The behaviour of high-dimensional measures is dominated by concentration of measure phenomena. The mode of a high-dimensional Gaussian is not where most of its mass lives; the mass concentrates on a thin shell. This single fact reframes a great deal of intuition about generative modelling, Bayesian inference, and adversarial examples, and it is rigorously a fact about measures. Engineering pedagogy can describe the consequences; measure-theoretic pedagogy explains them.

5. Anticipated Objections

5.1 "This is too abstract for practitioners."

This is the most common objection and the one we take most seriously. The response is twofold. First, the framework does not require students to absorb a graduate course in measure theory before they can train a model. It requires them to encounter the five foundational objects early and revisit them as their algorithmic vocabulary expands. The technical depth scales with the student's needs. Second, abstraction is not the opposite of practice; it is the substrate on which durable practice rests. A student who knows that KL requires absolute continuity can debug a NaN loss faster than a student who knows only that NaN losses sometimes occur.

5.2 "Most practitioners will never encounter measure-theoretic subtleties."

This was true a decade ago. It is not true now. Diffusion models are stochastic-analytic objects. Flow matching is an optimal-transport object. Score-based generative modelling is formulated through partial differential equations on measure spaces. The frontier has moved into territory where the engineering dialect is genuinely insufficient, and the rate at which it is moving is accelerating. A curriculum designed for the previous decade leaves students behind.

5.3 "There is no time in the curriculum for this."

The framework does not add time; it reorganises existing material. Probability is already taught in machine learning courses; we propose teaching it as the theory of measures rather than the theory of densities and PMFs. Optimisation is already taught; we propose connecting stochastic gradient descent to expectations under the empirical measure rather than treating it as a separate technique. The total content is comparable; the organisation is different.

5.4 "Engineers will never read measure-theoretic papers."

Engineers increasingly must. The implementation of a diffusion model requires choices about discretisation of the reverse SDE, conditioning mechanisms, and noise schedules that are correctly made only with reference to the underlying measure-theoretic structure. The implementation of a flow-based model requires understanding when the change-of-variables formula applies. We are not arguing that engineers must produce measure-theoretic research; we are arguing that they must read it competently, and the framework is what enables that reading.

5.5 "This privileges generative modelling over discriminative tasks."

The framework applies equally to discriminative tasks (Mode 2: conditional estimation), and arguably illuminates them more than the standard treatment. Cross-entropy as conditional KL, MSE as projection in a Hilbert space induced by P_X, importance weighting as

Radon-Nikodym derivatives, calibration as a property of the conditional measure — these are all standard topics rendered more rigorously. The framework's neutrality across modes is a feature.

6. A Curriculum Proposal

We close with a concrete proposal for a curriculum that teaches machine learning through this framework. The proposal is structured in three phases, suitable for a one-year graduate sequence or a two-year advanced undergraduate programme.

6.1 Phase One — The Foundational Objects (8–10 weeks)

Probability spaces, σ-algebras, and measures are introduced as the language for describing where data lives. Students learn the Radon-Nikodym theorem in a form that emphasises its operational role: the existence of likelihood ratios, importance weights, and conditional densities. The Lebesgue integral is introduced primarily as the integral that admits dominated and monotone convergence theorems, which are the analytic backbone of expectation, gradients of expectations, and SGD analysis.

By the end of Phase One, students can state precisely what the empirical measure is, what a pushforward measure is, and why the choice of divergence matters. They have not yet seen a neural network in this course.

6.2 Phase Two — The Operational Modes (10–12 weeks)

Each of the four modes is developed in turn, with one canonical algorithm fully worked out. Mode 1 is anchored by normalising flows (where the change-of-variables formula and the absolute continuity precondition are most visible). Mode 2 by logistic regression and its conditional KL interpretation. Mode 3 by WGAN, where the Kantorovich-Rubinstein duality connects the Wasserstein distance to the discriminator's Lipschitz constraint. Mode 4 by Langevin dynamics, where sampling from a measure is connected to its score function.

By the end of Phase Two, students can take a new algorithm from a paper and place it in the framework: identify its mode, its divergence, its regularity preconditions, and its likely failure modes.

6.3 Phase Three — Frontier and Synthesis (8–10 weeks)

Phase Three covers diffusion models (as time-reversed SDEs on path space), flow matching (as continuous-time optimal transport), score-based generative modelling (as Fokker-Planck dynamics), and the connections to optimal transport, ergodic theory, and information geometry. Students complete a final project that takes a recent paper and either reproduces its result, identifies a measure-theoretic precondition the paper does not state explicitly, or proposes a modification motivated by the framework.

6.4 Prerequisites and Supporting Material

The proposed sequence assumes real analysis at the level of Rudin's Principles, linear algebra at the level of Axler, and elementary probability. Measure theory itself is developed within the course; we do not require a prior measure theory course, though one would compress Phase One. Programming proficiency is assumed throughout, with PyTorch as the implementation vehicle.

6.5 Assessment

Assessment emphasises the predictions the framework licenses. Problem sets ask students to identify failure modes from preconditions, to derive divergence-specific gradient behaviour, and to map new algorithms onto the unified table of Section 3. The final project is the primary mechanism for assessing whether students can use the framework rather than merely recite it.

7. Position and Conclusion

The position this paper defends is that machine learning and deep learning, taught well, are taught as the theory of operations on probability measures. The five foundational objects (data measure, model measure, pushforward, divergence, Radon-Nikodym derivative) and the four operational modes (density estimation, conditional estimation, distribution matching, sampling) form a unified conceptual structure within which the major algorithm families of the field appear as instances rather than as separate inventions.

This is not a research programme — measure-theoretic foundations of machine learning are well-developed and have been for decades. It is a pedagogical programme. The claim is that what is missing from contemporary machine learning education is not new mathematics but a deliberate organisation of existing mathematics around the objects that contemporary research actually manipulates.

A student who completes a curriculum built on this framework can read a 2026 paper on flow matching, a 2025 paper on diffusion model conditioning, or a 2027 paper on whatever supersedes them, and place it in a structure they already understand. They can predict failure modes from preconditions. They can articulate why a divergence choice matters. They can connect their work to the mature mathematics on which the frontier increasingly draws.

The Erdos curriculum is well-positioned to develop this programme. The Mathematical Foundations of AI textbook already commits to a rigorous treatment of probability, generative models, and the trustworthy-AI considerations that follow from precise statements about measures. The proposal of this paper is to make the measure-theoretic spine of that treatment explicit, central, and the organising principle of how the field is taught.

The frontier has moved. The curriculum should move with it.

— END OF POSITION PAPER —